

Project: EduSphere - Digital Learning & Curriculum Management Platform

1. Introduction

EduSphere is a web-based Digital Learning and Curriculum Management Platform designed for schools, universities, and corporate training programs to unify course delivery, student engagement, and assessment tracking. The platform consolidates curriculum data, assignments, attendance, and performance metrics to provide end-to-end visibility, exception handling, and proactive notifications.

It supports RESTful APIs with Angular/React frontends, and backend implementations in Java Spring Boot or ASP.NET Core. The architecture emphasizes scalability, secure handling of student data, and compliance with educational standards.

Actors / Users

- **Student:** Accesses courses, submits assignments, tracks progress.
- **Teacher / Instructor:** Creates courses, manages content, grades assignments.
- **Academic Coordinator:** Oversees curriculum, schedules, and compliance.
- **Examiner:** Manages assessments, results, and reports.
- **Parent / Guardian:** Views student progress and attendance.
- **Admin:** Configures courses, roles, users, and integrations.

2. Module Overview

2.1 Identity & Access Management

Authentication, RBAC, audit trails.

2.2 Curriculum & Course Management

Course creation, syllabus management, and scheduling.

2.3 Student Enrollment & Attendance

Enrollment workflows, attendance tracking, and reporting.

2.4 Assignment & Assessment Management

Assignment submission, grading, and exam scheduling.

2.5 Performance & Progress Tracking

Student dashboards, progress reports, and KPIs.

2.6 Analytics & Reporting

Academic dashboards, performance trends, and compliance reports.

2.7 Notifications & Alerts

Real-time alerts for assignments, exams, and attendance.

3. Architecture Overview

- **Frontend:** Angular or React SPA for dashboards and workflows.
- **Backend:** RESTful microservices (Spring Boot / ASP.NET Core).
- **Database:** Relational DB (Postgres/MySQL/SQL Server).
- **Eventing:** Kafka or cloud pub/sub for assignment and exam events.
- **Cache:** Redis for session state and hot lookups.
- **Integration:** ERP, HR, and external learning content providers.
- **Security:** TLS, encryption at rest, RBAC, audit logging.

4. Module Wise Design

4.1 Identity & Access Management Module

Features

- User provisioning and role assignment
- Secure login with RBAC and SSO support
- Session management and password policies
- Immutable audit logs for compliance

Entities

- **User**(UserID, Name, Role, Email, Phone)
- **AuditLog**(AuditID, UserID, Action, Timestamp)

4.2 Curriculum & Course Management Module

Features

- Course creation and editing
- Syllabus management with topic breakdown
- Scheduling and calendar integration
- Course lifecycle management (Active/Inactive)

Entities

- **Course**(CourseID, Title, Description, Credits, Status)
- **Syllabus**(SyllabusID, CourseID, Topics, Duration, EffectiveDate)

4.3 Student Enrollment & Attendance Module

Features

- Student enrollment workflows
- Attendance tracking per course/session

- Automated attendance reports
- Integration with student information systems

Entities

- **Enrollment**(EnrollmentID, StudentID, CourseID, EnrollmentDate, Status)
- **Attendance**(AttendanceID, StudentID, CourseID, Date, Status)

4.4 Assignment & Assessment Management Module

Features

- Assignment creation and submission tracking
- Grading workflows with feedback
- Exam scheduling and monitoring
- Result publishing and archival

Entities

- **Assignment**(AssignmentID, CourseID, Title, DueDate, Status)
- **Submission**(SubmissionID, AssignmentID, StudentID, SubmittedDate, Grade)
- **Exam**(ExamID, CourseID, ExamDate, Type, Status)

4.5 Performance & Progress Tracking Module

Features

- Student progress dashboards
- KPI tracking (grades, attendance, completion)
- Automated progress report generation
- Parent/guardian visibility into performance

Entities

- **ProgressReport**(ReportID, StudentID, CourseID, Metrics, GeneratedDate)

4.6 Analytics & Reporting Module

Features

- Academic dashboards for teachers and coordinators
- Performance trend analysis
- Attendance and pass rate statistics
- Compliance reporting for accreditation

Entities

- **KPIReport**(ReportID, Scope, Metrics, GeneratedDate)

4.7 Notifications & Alerts Module

Features

- Real-time alerts for assignments, exams, and attendance
- Escalation workflows for missed deadlines
- Configurable notification channels (in-app, email, SMS)
- Notification history and audit

Entities

- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

5. Deployment Strategy

- **Local/Dev:** Angular/React UI, Spring Boot/.NET backend, local DB.
- **Staging:** Containerized services with synthetic academic data.
- **Production:** Kubernetes orchestration, API gateway, managed RDBMS with replicas, Kafka for events, Redis for caching.
- **High Availability:** Multi-AZ deployments, auto-scaling, disaster recovery playbooks.

6. Database Design

Tables

- **User**(UserID, Name, Role, Email, Phone)
- **AuditLog**(AuditID, UserID, Action, Timestamp)
- **Course**(CourseID, Title, Description, Credits, Status)
- **Syllabus**(SyllabusID, CourseID, Topics, Duration, EffectiveDate)
- **Enrollment**(EnrollmentID, StudentID, CourseID, EnrollmentDate, Status)
- **Attendance**(AttendanceID, StudentID, CourseID, Date, Status)
- **Assignment**(AssignmentID, CourseID, Title, DueDate, Status)
- **Submission**(SubmissionID, AssignmentID, StudentID, SubmittedDate, Grade)
- **Exam**(ExamID, CourseID, ExamDate, Type, Status)
- **ProgressReport**(ReportID, StudentID, CourseID, Metrics, GeneratedDate)
- **KPIReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Student Dashboard:** Courses, assignments, exams, progress.
- **Teacher Console:** Course creation, grading, attendance.
- **Coordinator Dashboard:** Curriculum overview, compliance, scheduling.
- **Examiner Console:** Exam scheduling, grading, reports.
- **Parent Portal:** Student progress, attendance, notifications.
- **Admin Console:** Course setup, user roles, integrations.

8. Non-Functional Requirements

- **Performance:** Handle up to **50,000 concurrent students**; assignment submission latency under **300 ms**.
- **Security:** RBAC, TLS, encryption at rest, GDPR compliance, immutable audit logs.
- **Scalability:** Horizontal scaling for enrollment and reporting services; multi-tenant support.
- **Availability:** Target **99.9% uptime** for academic services.
- **Maintainability:** Modular APIs, versioning, automated migrations, CI/CD pipelines.
- **Observability:** Centralized logs, metrics, tracing, and alerting.
- **Compliance:** Configurable retention policies, audit trails, regulatory reporting formats.

9. Assumptions & Constraints

- **Phase 1 scope:** Course management, enrollment, attendance, assignments, exams, and dashboards.
- **Excluded in Phase 1:** Advanced AI-based adaptive learning, gamification features, and external certification integrations.
- **Data availability:** Historical academic data available for reporting and calibration.
- **Integration:** ERP and external content providers via APIs.
- **Implementable:** Using Java Spring Boot or ASP.NET Core, Angular/React, Kafka, Redis, and relational DB tooling.